Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

MT614 – LINEAR ALGEBRA

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Prove that L(S) is a subspace of V.
2. Prove that $||\alpha\ u||=|\alpha|\ ||u||$.
3. Define Right invertible.
4. When do you say that two linear transformations S,T $\in A(V)$ are similar.
5. For $A, B \in F_n$ and $\lambda \in F$ prove that $tr(\lambda A) = \lambda\ tr\ A$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. IF $v_1, v_2 ... v_n$ are in V then prove that either they are linearly independent or some v_k is a linear combination of the preceeding vectors $v_1, v_2 ... v_{k-1}$.
7. If $u, v \in V$ then prove that $|(u,v)| \le \|u\|\|v\|$.
8. If V is finite-dimensional over F, then prove that $T \in A(V)$ is invertible if and only if the constant term of the minimal polynomial for T is not 0.
9. Suppose that V is the vector space of all polynomials over F of degree 3 or less, and let be the differentiation operator defined $(\alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3)$ $D = \alpha_1 + 2\alpha_2 x + 3\alpha_3 x^2$. *In* the basis $v_1 = 1, v_2 = x, v_3 = x^2, v_4 = x^3$ find the matrix of D.
10. If F is of characteristic 0 and if S and T, in $A_F(V)$, are such ST – TS commutes with S, then prove that ST – TS is nilpotent.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) If V is finite-dimensional and if W is a subspace of V then prove that W is finite dimensional, dim W ≤ dim V and $\dim \frac{V}{W} = \dim V - \dim W$

(or)

b) If V is finite-dimensional over F then prove that any two bases of V have the same number of elements.

12. a) If V is finite-dimensional and W is a subspace of V, then prove that W is isomorphic to $\frac{V}{A(W)}$ and dim A(W) = dim V – dim W.

(or)

b) Let V be a finite-dimensional inner product space; then prove that V has an orthonormal set as a basis.

13. a) (i) If $\lambda \in F$ is a characteristic root of $T \in A(V)$ then prove that for any polynomial $q(x) \in F[x]$ $q(\lambda)$ is a characteristic root of q(T).

(ii) If $\lambda \in F$ is a characteristic root of $T \in A(V)$ then prove that λ is a root of the minimal polynomial of T. and also prove that T only has a finite number of characteristic roots in F.

(or)

b) If $\lambda_1,....\lambda_k$ in F are distinct characteristic roots of $T \in A(V)$ and if $v_1,....,v_k$ are characteristic vectors of T belonging to $\lambda_1,....\lambda_k$ respectively, then prove that $v_1,....,v_k$ are linearly independent over F.

14. a) If V is n-dimensional over F and if $T \in A(V)$ has the matrix m_1(T) in the basis $v_1.....v_n$ and the matrix m_2(T) in the basis w_1, …..w_n of V over F, then prove that there is an element $C \in F_n$ such that $m_2(T) = Cm_1(T)C^{-1}$

(or)

b) If V is n-dimensional over F and if $T \in A(V)$ has all its characteristic roots in F, then prove that T satisfies a polynomial of degree n over F.

15. a) If F is a field of characteristic 0, and if $T \in A_F(V)$ is such that tr $T^i = 0$ for all $i \geq 1$ then prove that T is nilpotent.

(or)

b) (i). If two rows of A are equal (that is, $v_r = v_s$ for $r \neq s$), then prove that det A = 0.
(ii). Prove that interchanging two rows of A changes the sign of its determinant.

****